

CALEB SKINNER

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2727 Revere St Apt 3014 Houston, TX 77098

EDUCATION

Rice University – Houston, TX

- PhD in Statistics (M.S. earned en route) (ongoing)
- Advisor: Dr. Meng Li
- GPA: 3.97
- Qualifying Exam - Passed Summer 2025

Baylor University – Waco, TX

- Bachelor of Business Administration May 2024
- Majors: Statistics, Economics, Mathematics, Business Fellows
- Honors Program; Overall GPA: 3.97

WORKING PAPERS

Skinner, C, Li, M. "Exact Partial Constancy Prior for Structured Ensemble Forecasting"

- Proposes a widely applicable exact partial constancy prior for a principled Bayesian approach to forecast aggregation and ensemble learning.

Skinner, C, Guo, Y. & Li, M. "LLM Sparsity Prior for Robust Feature Selection"

- Introduces a novel prior integrating feature importance weights derived from large language models into classic Bayesian Variable Selection techniques like Spike-and-Slab and Spike-and-Slab Lasso.

PUBLICATIONS

Skinner, C, Patrick, J, Bryson, C. & Sturdivant, R. (2025, June 16). Mixed Martial Arts - Interrater Reliability. "The SCORE Network." <https://doi.org/10.17605/OSF.IO/F8VBQ>

- Interactive teaching tool details Interrater Reliability using data from Mixed Martial Arts judges.

Skinner, C, Sturdivant, R., "Is Momentum Real: On the Existence and Influences of Psychological Momentum in Tennis" (Honors Undergraduate Thesis)

- Defining momentum as the backward approximation of the derivative of the players' smoothed point margin curve, we find that a player's momentum entering a point has a significant impact on his or her likelihood of winning the next point.

HONORS AND AWARDS

- Getterman Scholar – Full Ride Scholarship to Baylor University 2020 – 2024
- Crane Scholar (Highly Selective Interdisciplinary Program) 2021 – 2024
- Most Outstanding Undergraduate in Statistics May 2024

OTHER RESEARCH PROJECTS

Sleeper Bayesian

Fall 2024

- Developed a Normal Bayesian Hierarchical Model to estimate NBA player performance and future outcomes. Incorporated Game Theory decision boundaries to optimize fantasy basketball roster decisions.

<i>Sparse Trustworthy Variational Bayes</i>	Spring 2025
Implemented a novel extension of Trustworthy Variational Bayes (Liu and Li, 2024) applied to the discrete spike-and-slab prior (Ray and Szabo, 2022) to address variance underestimation and improve variational credible intervals coverage.	
<i>Model Averaged Doubly Robust Estimator</i>	Spring 2025
Compared the Model Averaged Doubly Robust Estimator (Cefalu et al., 2017) with competing methods like stepwise model selection and Bayesian Adjustment for Confounding (Wang et al., 2012) in a three-part simulation study.	
<i>Detecting Presence of Brain Aneurysms</i>	Fall 2025
Develop a deep learning pipeline for identification and localization of brain aneurysms from three-dimensional brain scans. Pre-processed multi-modal scans and implemented a two-step 3-D Convolutional Neural Network with modern block architecture.	

STATISTICAL EXPERIENCES

<i>Data to Knowledge Lab Mentor</i>	Spring 2026
Mentored a team of six undergraduates twice a week in applying text clustering, statistical inference, and dashboard development to analyze work orders for a facilities management client.	
<i>Rice Teaching Assistant</i>	August 2024 – Present
Led study sessions and office hours for undergraduate statistics; responsible for developing exam questions and grading assignments.	
<i>SCORE Network Material Developer</i>	May 2023 – August 2023
- Developed three extensive (20-50 page) interactive lesson plans communicating statistical concepts (Cluster Analysis, Interrater Reliability, Logistic Regression). - Produced interactive and engaging modules utilizing R-Markdown and ISLE.	
<i>Sports Analytics Club – Vice President</i>	August 2021 – May 2024
Developed a production-ready R Shiny application, delivering numerous interactive visualizations and predictive tables utilized by baseball coaching staff.	
<i>Undergraduate Economics Research Assistant</i>	September 2023 – May 2024
Compiled a data set for a meta-research study under Dr. Mark Hoekstra.	

MEMBERSHIPS

<i>American Statistical Association`</i>	2024-Present
<i>Section on Bayesian Statistical Science</i>	2025-Present
<i>Business and Economic Statistics Section</i>	2025-Present

VOLUNTEERING

<i>Vice President – Rice ASA Student Chapter</i>	2025-Present
<i>Judge for Undergraduate Statistics Capstone</i>	Fall 2025, Spring 2026
<i>Prospective Student Preview Day - PhD Coordinator</i>	2026
<i>Prospective Student Preview Day - PhD Representative</i>	2025

MISCELLANEOUS

<i>Mathematical Contest in Modeling (MCM), Problem C Writer</i>	2023
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